CSS 426 MLOPS

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ROLL NUMBER : 2022BCS0001

BATCH : 1

ASSIGNMENT NO. : 1

PART : 2

GITHUB LINK : https://github.com/Vishnu0107/2022BCS0001_churn_system

DOCKERHUB LINK : https://hub.docker.com/repository/docker/centronox/churn-api/general

1. Introduction

This project implements a **rule-based customer churn prediction system** using a **DevOps pipeline**. The system uses deterministic rules instead of ML, ensuring transparency, consistency, and ease of validation.

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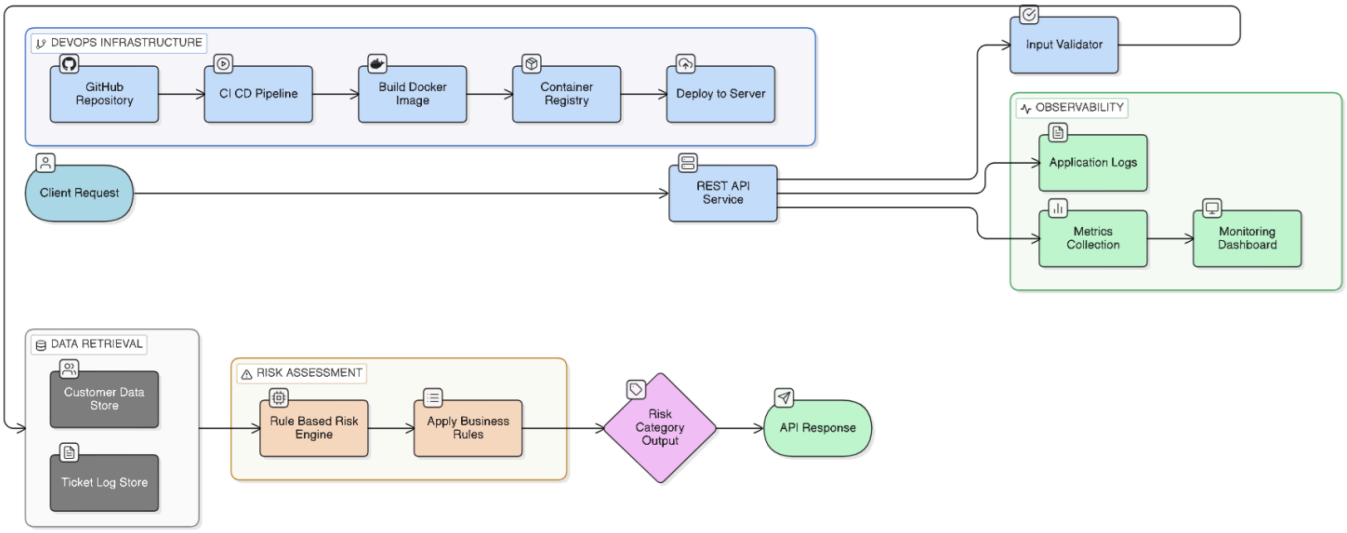
2. DevOps Workflow



Workflow:

- Code pushed to GitHub
- CI pipeline triggered
- Tests executed
- Docker image built
- Image pushed to DockerHub

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3. System Overview

Components:

- FastAPI REST API
- Rule-based prediction engine
- Input validation
- Database (SQLite/PostgreSQL)
- GitHub Actions (CI/CD)
- Docker container

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4. Prediction Logic

```
def evaluate_risk(data):
if data["tenure"] < 6 and data["MonthlyCharges"] > 70:
return "High Risk"
elif data["tenure"] < 12:
return "Medium Risk"
else:
return "Low Risk"
```

5. Input Validation

```
from fastapi import HTTPException

def validate_input(data):
    if "tenure" not in data:
        raise HTTPException(status_code=400, detail="tenure missing")
    if "MonthlyCharges" not in data:
```

```
        raise HTTPException(status_code=400, detail="MonthlyCharges missing")
    return data
```

6. API Endpoint

```
@app.post("/predict")
def predict(data: dict, db=Depends(get_db)):
    data = validate_input(data)
    risk = evaluate_risk(data)
    return {"risk": risk}
```

Sample input and output

```


INPUT :
{"tenure": 5, "MonthlyCharges": 80}

OUTPUT :
{"risk": "High Risk"}
```

7. Testing

Tests validate:

- API response
- Rule correctness
- Input validation

8. CI/CD Pipeline

```
name: CI/CD Pipeline

on:
  push:
    branches: ["main"]

jobs:
  build-test-deploy:
    runs-on: ubuntu-latest

    steps:
    - name: Checkout Code
      uses: actions/checkout@v4
```

```yaml
    - name: Set up Python
      uses: actions/setup-python@v5
      with:
        python-version: "3.10"

    - name: Install dependencies
      run: pip install -r requirements.txt

    - name: Set environment variables
      run: |
        echo "DATABASE_URL=sqlite:///./test.db" >> $GITHUB_ENV
        echo "PYTHONPATH=$PWD" >> $GITHUB_ENV

    - name: Run Tests
      run: |
        pytest --maxfail=1 --disable-warnings -q --tb=short | tee
pytest_output.txt

    - name: Upload Test Results
      uses: actions/upload-artifact@v4
      with:
        name: pytest-results
        path: pytest_output.txt

    - name: Build Docker Image
      run: docker build -t ${{ secrets.DOCKERHUB_USERNAME }}/churn-api:latest .

    - name: Login to DockerHub
      run: echo "${{ secrets.DOCKERHUB_TOKEN }}" | docker login -u "${{
secrets.DOCKERHUB_USERNAME }}" --password-stdin

    - name: Push Docker Image
      run: docker push ${{ secrets.DOCKERHUB_USERNAME }}/churn-api:latest

    - name: Summary
      run: |
        echo "## CI/CD Pipeline Report" >> $GITHUB_STEP_SUMMARY

        echo "### Build Information" >> $GITHUB_STEP_SUMMARY
        echo "- Trigger: Push to main branch" >> $GITHUB_STEP_SUMMARY
        echo "- Environment: Ubuntu Latest (GitHub Actions Runner)" >>
$GITHUB_STEP_SUMMARY

        echo "" >> $GITHUB_STEP_SUMMARY
        echo "### Test Execution" >> $GITHUB_STEP_SUMMARY
```

```
        echo "- Test Framework: Pytest" >> $GITHUB_STEP_SUMMARY
        echo "- Status: All test cases passed successfully" >>
$GITHUB_STEP_SUMMARY
        echo "- API endpoint (/predict) validated" >> $GITHUB_STEP_SUMMARY
        echo "- Business rule logic verified" >> $GITHUB_STEP_SUMMARY

        echo "" >> $GITHUB_STEP_SUMMARY
        echo "### Containerization" >> $GITHUB_STEP_SUMMARY
        echo "- Docker image built successfully" >> $GITHUB_STEP_SUMMARY
        echo "- Image tagged using DockerHub repository" >> $GITHUB_STEP_SUMMARY

        echo "" >> $GITHUB_STEP_SUMMARY
        echo "### Deployment Readiness" >> $GITHUB_STEP_SUMMARY
        echo "- Image pushed to DockerHub" >> $GITHUB_STEP_SUMMARY
        echo "- Ready for containerized deployment" >> $GITHUB_STEP_SUMMARY

        echo "" >> $GITHUB_STEP_SUMMARY
        echo "### Overall Status" >> $GITHUB_STEP_SUMMARY
        echo "Pipeline completed successfully with all stages executed without
errors." >> $GITHUB_STEP_SUMMARY
```

Steps:

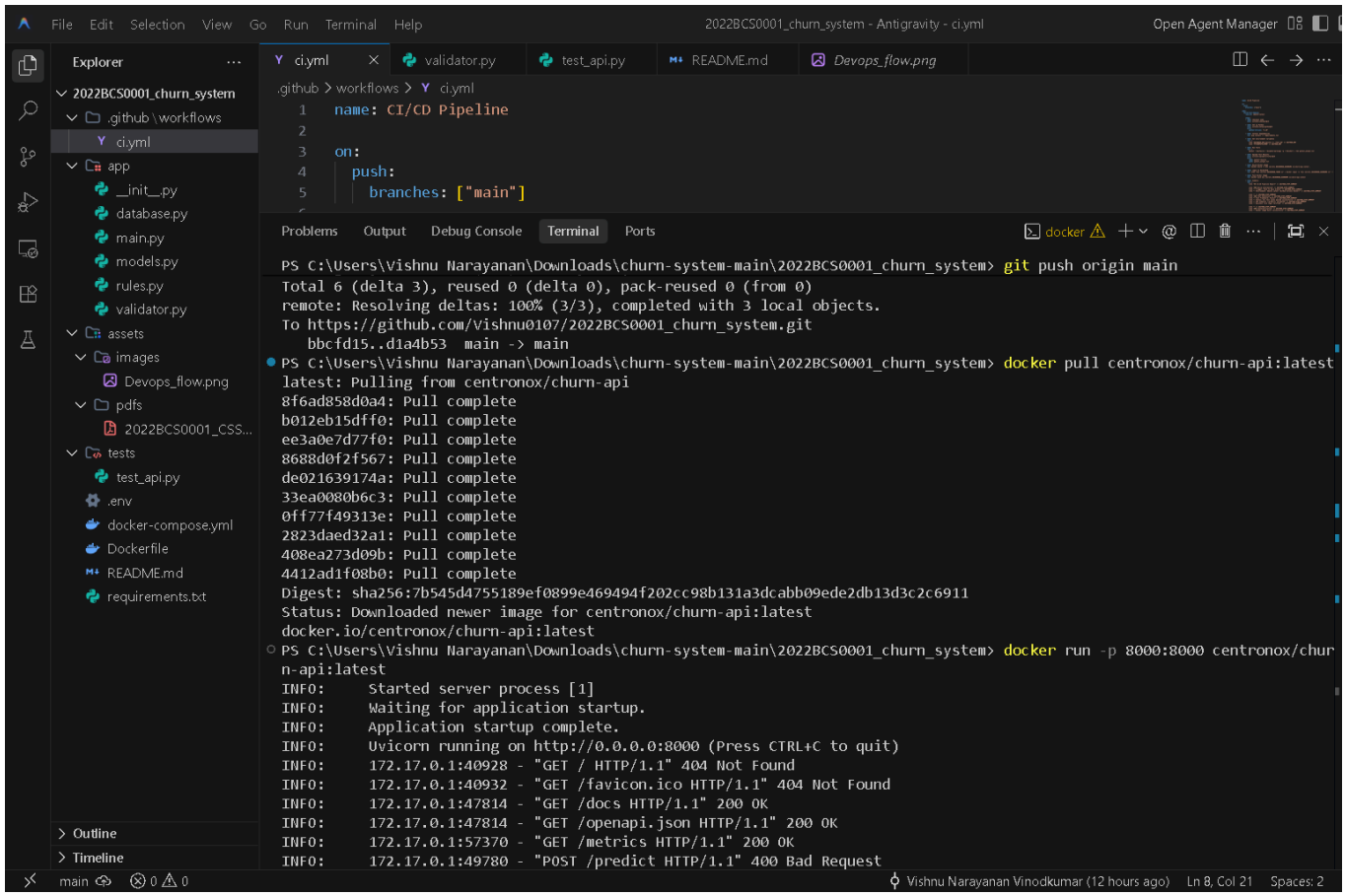
- Install dependencies
- Run pytest
- Build Docker image
- Push to DockerHub
- Made it into a docker repository
- Deployed it

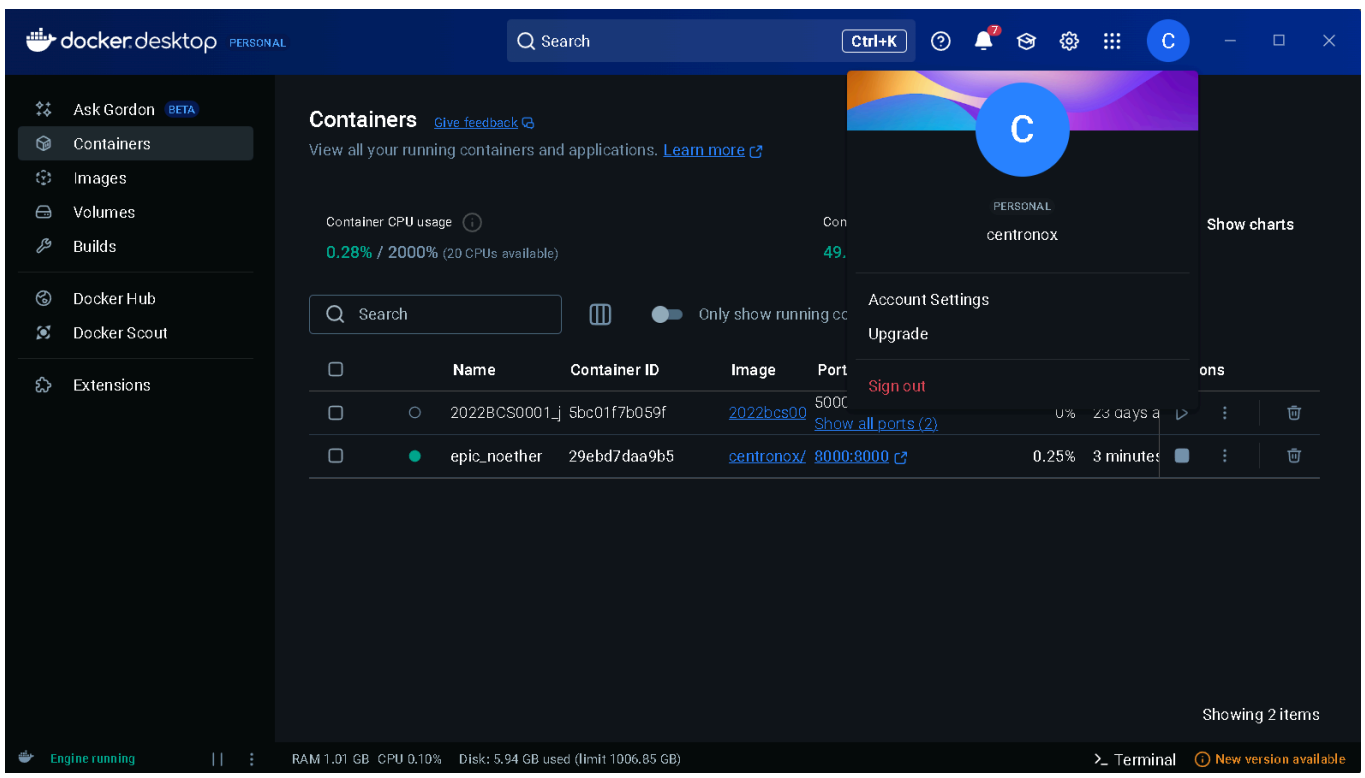
9. Conclusion

The project demonstrates a complete DevOps workflow including testing, containerization, and CI/CD integration. The system is reliable, reproducible, and deployment-ready.

10. Snippets

Running Locally


File Edit Selection View Go Run Terminal Help
2022BCS0001_churn_system - Antigravity - ci.yml
Open Agent Manager
Explorer
2022BCS0001_churn_system
.github \ workflows
ci.yml
app
__init__.py
database.py
main.py
models.py
rules.py
validator.py
assets
images
Devops_flow.png
pdfs
2022BCS0001_CSS...
tests
test_api.py
.env
docker-compose.yml
Dockerfile
README.md
requirements.txt
Outline
Timeline
ci.yml
validator.py
test_api.py
README.md
Devops_flow.png
.github > workflows > ci.yml
name: CI/CD Pipeline
on:
push:
branches: ["main"]
Problems Output Debug Console Terminal Ports
docker
PS C:\Users\Vishnu Narayanan\Downloads\churn-system-main\2022BCS0001_churn_system> git push origin main
Total 6 (delta 3), reused 0 (delta 0), pack-reused 0 (from 0)
remote: Resolving deltas: 100% (3/3), completed with 3 local objects.
To https://github.com/Vishnu0107/2022BCS0001_churn_system.git
bbcfd15..d1a4b53 main -> main
PS C:\Users\Vishnu Narayanan\Downloads\churn-system-main\2022BCS0001_churn_system> docker pull centronox/churn-api:latest
latest: Pulling from centronox/churn-api
8f6ad858d0a4: Pull complete
b012eb15dff0: Pull complete
ee3a0e7d77f0: Pull complete
8688d0f2f567: Pull complete
de021639174a: Pull complete
33ea0080b6c3: Pull complete
0ff77f49313e: Pull complete
2823daed32a1: Pull complete
408ea273d09b: Pull complete
4412ad1f08b0: Pull complete
Digest: sha256:7b545d4755189ef0899e469494f202cc98b131a3dcabb09ede2db13d3c2c6911
Status: Downloaded newer image for centronox/churn-api:latest
docker.io/centronox/churn-api:latest
PS C:\Users\Vishnu Narayanan\Downloads\churn-system-main\2022BCS0001_churn_system> docker run -p 8000:8000 centronox/churn-api:latest
INFO: Started server process [1]
INFO: Waiting for application startup.
INFO: Application startup complete.
INFO: Uvicorn running on http://0.0.0.0:8000 (Press CTRL+C to quit)
INFO: 172.17.0.1:40928 - "GET / HTTP/1.1" 404 Not Found
INFO: 172.17.0.1:40932 - "GET /favicon.ico HTTP/1.1" 404 Not Found
INFO: 172.17.0.1:47814 - "GET /docs HTTP/1.1" 200 OK
INFO: 172.17.0.1:47814 - "GET /openapi.json HTTP/1.1" 200 OK
INFO: 172.17.0.1:57370 - "GET /metrics HTTP/1.1" 200 OK
INFO: 172.17.0.1:49780 - "POST /predict HTTP/1.1" 400 Bad Request
main
Vishnu Narayanan Vinodkumar (12 hours ago)
Ln 8, Col 21
Spaces: 2


docker desktop PERSONAL
Search
Ctrl+K
Ask Gordon BETA
Containers
Images
Volumes
Builds
Docker Hub
Docker Scout
Extensions
Containers
Give feedback
View all your running containers and applications. Learn more
Container CPU usage
0.28% / 2000% (20 CPUs available)
Show charts
Search
Only show running co
Name
Container ID
Image
Port
2022BCS0001_j
5bc01f7b059f
2022bcs00
Show all ports (2)
epic_noether
29ebd7daa9b5
centronox/
8000:8000
0.25%
3 minutes
Showing 2 items
PERSONAL
centronox
Account Settings
Upgrade
Sign out
Engine running
RAM 1.01 GB CPU 0.10% Disk: 5.94 GB used (limit 1006.85 GB)
Terminal
New version available

hub.docker.com/repository/docker/centronox/churn-api/general

Interview | Anime_&_Movies | Dashboard | Mails | Honors & BTP | Spotting Errors Sectio...

hub | Explore | My Hub | Search Docker Hub | CtrlK

centronox
Docker Personal

- Repositories
- Hardened Images
- Collaborations
- Settings
  - Default privacy
  - Notifications
- Billing
- Usage
  - Pulls
  - Storage

Upgrade plan

Repositories / churn-api / General

Using 0 of 1 private repositories.

centronox/churn-api

Last pushed 12 minutes ago • Repository size: 499 MB • 0 • 27

Add a description

Add a category

Docker commands | Public view

To push a new tag to this repository:

```
docker push centronox/churn-api:tagname
```

General | Tags | Image Management | Collaborators | Webhooks | Settings

Tags

DOCKER SCOUT INACTIVE
Activate

This repository contains 1 tag(s).

Tag	OS	Type	Pulled	Pushed
latest		Image	less than 1 day	13 minutes

See all

Repository overview

INCOMPLETE

An overview describes what your image does and how to run it. It displays in the

buildcloud

Build with
Docker Build Cloud

Accelerate image build times with access to cloud-based builders and shared cache.

Docker Build Cloud executes builds on optimally-dimensioned cloud infrastructure with dedicated per-organization isolation.

Get faster builds through shared caching across your team, native multi-platform support, and encrypted data transfer - all without managing infrastructure.

http://localhost:8000/docs#/

Interview | Anime_&_Movies | Dashboard | Mails | Honors & BTP | Spotting Errors Sectio...

FastAPI 0.1.0 OAS 3.1

/openapi.json

default

GET /metrics Metrics

POST /predict Predict

Parameters | Cancel | Reset

No parameters

Request body required | application/json

Edit Value | Schema

```
{
  "tenure": 5,
  "MonthlyCharges": 80
}
```

Code	Details
200	Response body

```
{
  "risk": "High Risk"
}
```

Download

Response headers

```
content-length: 20
content-type: application/json
date: Sat,21 Mar 2026 18:46:00 GMT
server: uvicorn
```

Responses

Code	Description	Links
200	Successful Response	*No links*

Media type

application/json

Controls Accept header.

Request body required

application/json

Edit Value | Schema

```
{
  "tenure": 10,
  "MonthlyCharges": 50
}
```

Execute | Clear

Responses

Curl

```
curl -X 'POST' \
  'http://localhost:8000/predict' \
  -H 'accept: application/json' \
  -H 'Content-Type: application/json' \
  -d '{
  "tenure": 10,
  "MonthlyCharges": 50
}'
```

Request URL

http://localhost:8000/predict

Server response

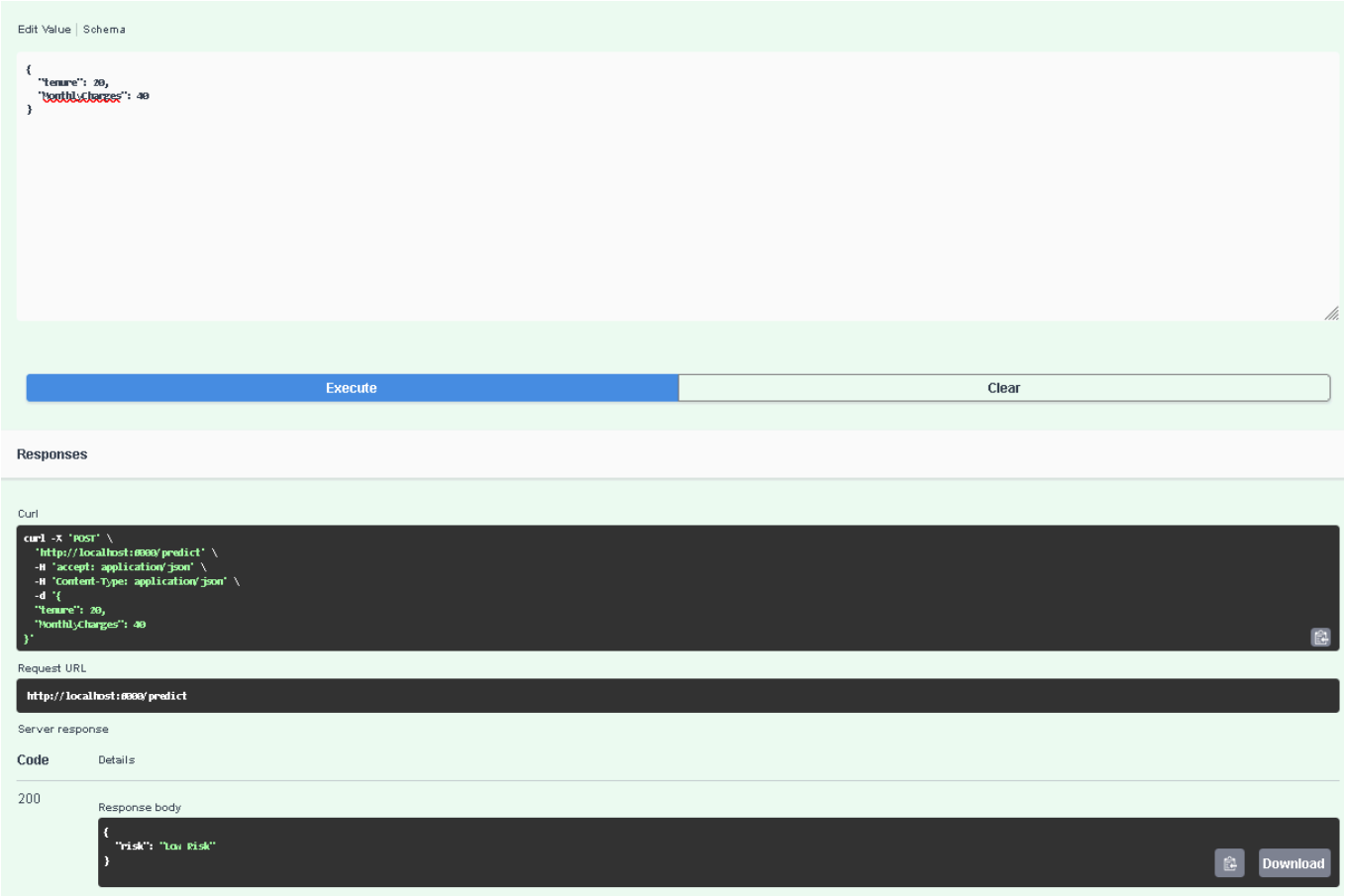
Code	Details
200	Response body

```
{
  "risk": "Medium Risk"
}
```

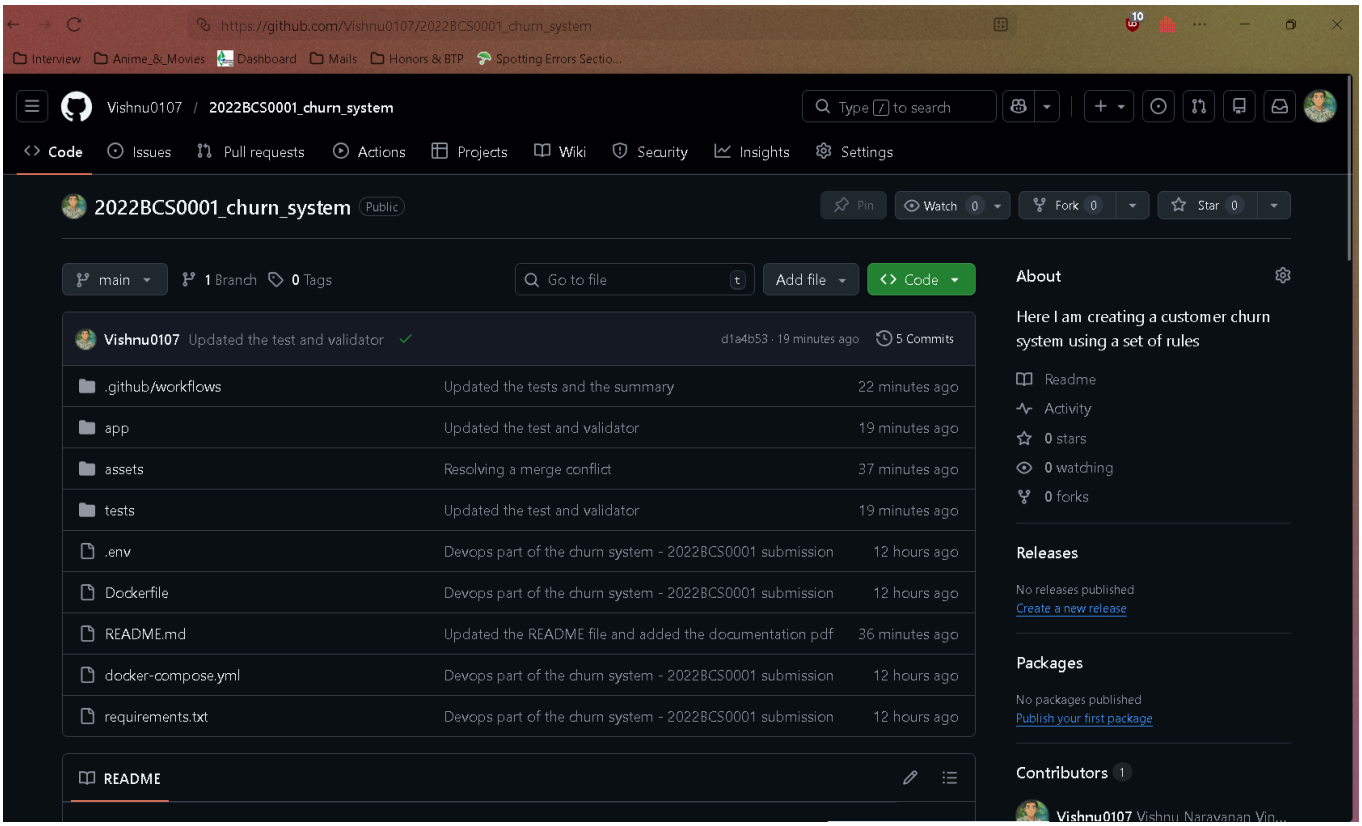
Download

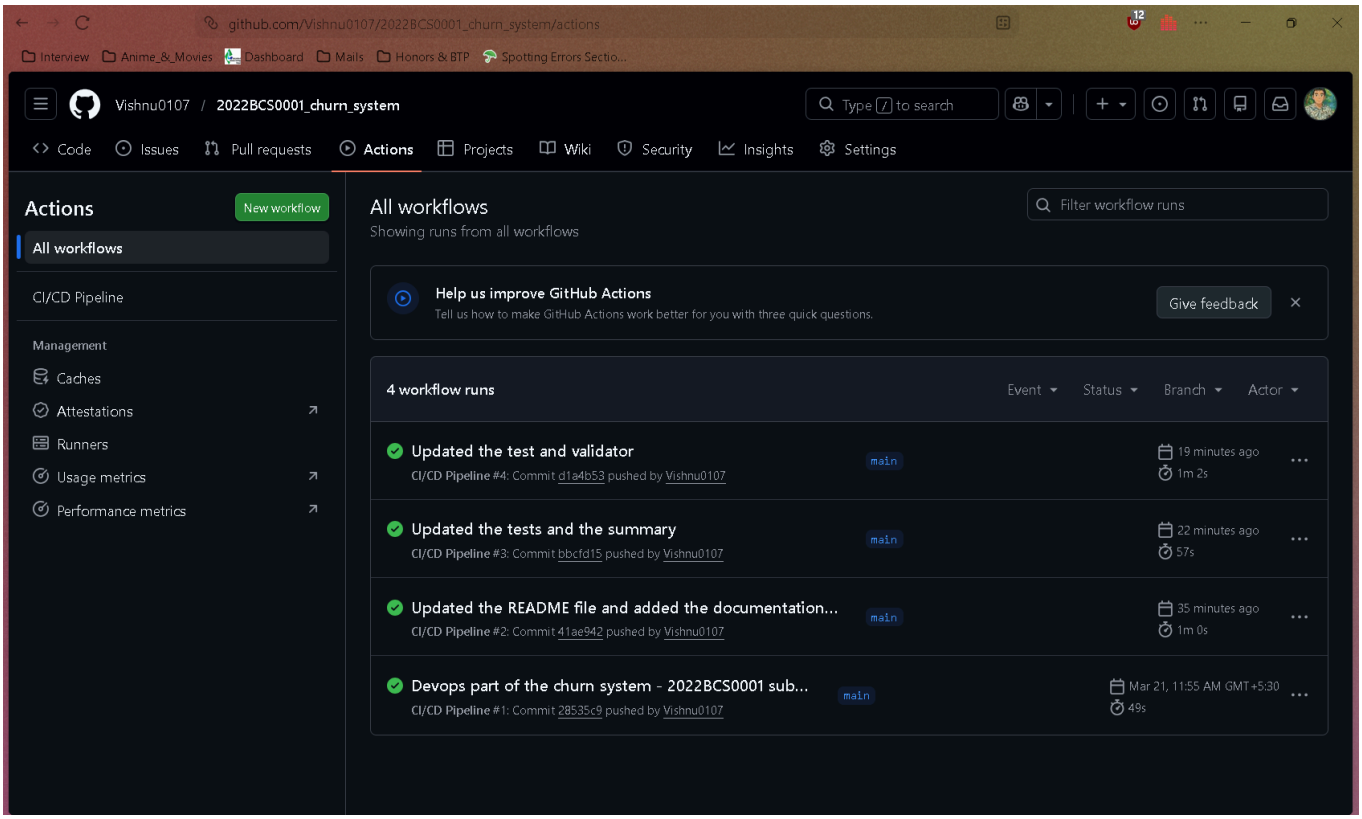
Response headers

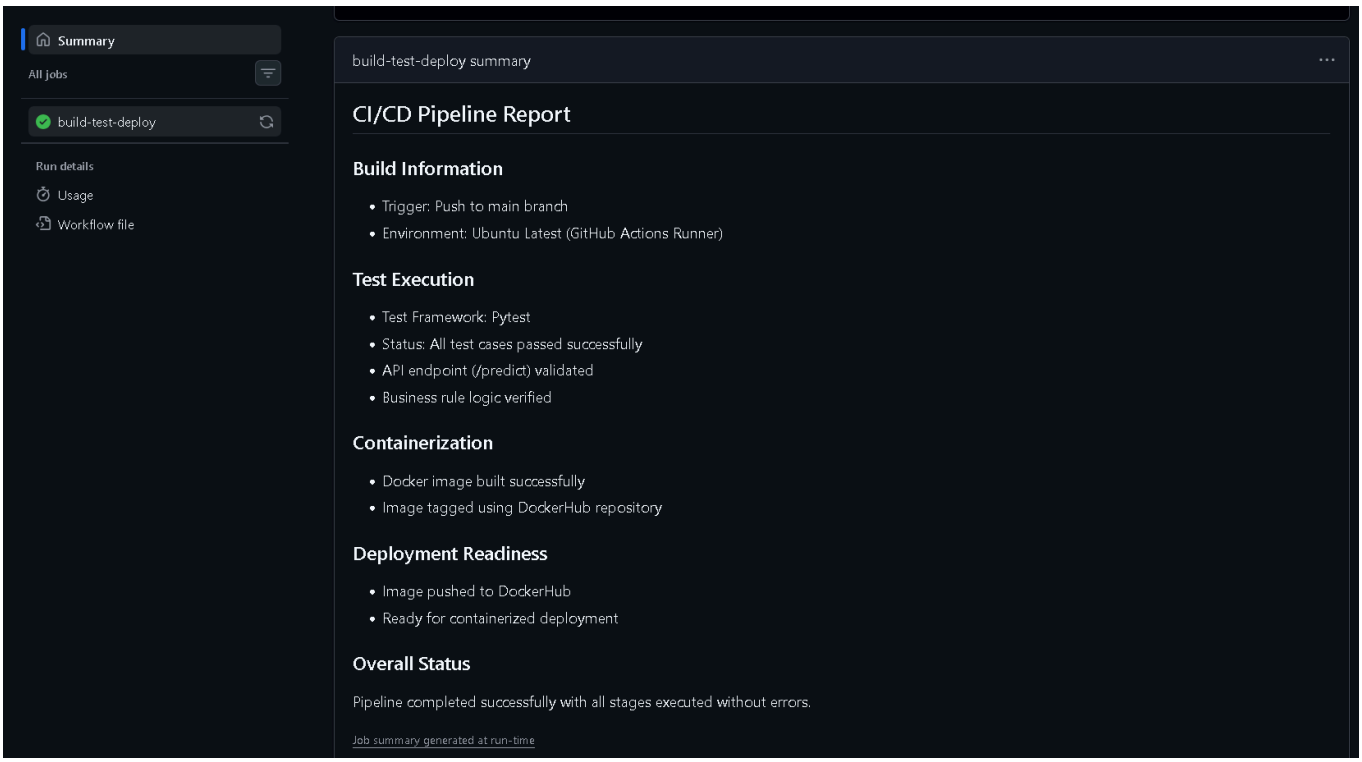
```
content-length: 22
content-type: application/json
date: Sat,21 Mar 2026 18:48:01 GMT
server: uvicorn
```


Edit Value | Schema
{
"tenure": 20,
"MonthlyCharges": 40
}
Execute
Clear
Responses
Curl
curl -X 'POST' \
'http://localhost:8000/predict' \
-H 'accept: application/json' \
-H 'Content-Type: application/json' \
-d '{
"tenure": 20,
"MonthlyCharges": 40
}'
Request URL
http://localhost:8000/predict
Server response
Code
Details
200
Response body
{
"risk": "Low Risk"
}
Download

Github Flow


https://github.com/Vishnu0107/2022BCS0001_churn_system
Vishnu0107 / 2022BCS0001_churn_system
Code
Issues
Pull requests
Actions
Projects
Wiki
Security
Insights
Settings
2022BCS0001_churn_system
Public
Watch 0
Fork 0
Star 0
main
1 Branch
0 Tags
Go to file
Add file
Code
Vishnu0107 Updated the test and validator
d1a4b53 · 19 minutes ago
5 Commits
.github/workflows
Updated the tests and the summary
22 minutes ago
app
Updated the test and validator
19 minutes ago
assets
Resolving a merge conflict
37 minutes ago
tests
Updated the test and validator
19 minutes ago
.env
Devops part of the churn system - 2022BCS0001 submission
12 hours ago
Dockerfile
Devops part of the churn system - 2022BCS0001 submission
12 hours ago
README.md
Updated the README file and added the documentation pdf
36 minutes ago
docker-compose.yml
Devops part of the churn system - 2022BCS0001 submission
12 hours ago
requirements.txt
Devops part of the churn system - 2022BCS0001 submission
12 hours ago
README
About
Here I am creating a customer churn system using a set of rules
Readme
Activity
0 stars
0 watching
0 forks
Releases
No releases published
Create a new release
Packages
No packages published
Publish your first package
Contributors 1


github.com/Vishnu0107/2022BCS0001_churn_system/actions
Vishnu0107 / 2022BCS0001_churn_system
Code
Issues
Pull requests
Actions
Projects
Wiki
Security
Insights
Settings
Actions
New workflow
All workflows
CI/CD Pipeline
Management
Caches
Attestations
Runners
Usage metrics
Performance metrics
All workflows
Showing runs from all workflows
Filter workflow runs
Help us improve GitHub Actions
Tell us how to make GitHub Actions work better for you with three quick questions.
Give feedback
4 workflow runs
Event
Status
Branch
Actor
Updated the test and validator
CI/CD Pipeline #4: Commit d1a4b53 pushed by Vishnu0107
main
19 minutes ago
1m 2s
Updated the tests and the summary
CI/CD Pipeline #3: Commit bbcfd15 pushed by Vishnu0107
main
22 minutes ago
57s
Updated the README file and added the documentation...
CI/CD Pipeline #2: Commit 41ae942 pushed by Vishnu0107
main
35 minutes ago
1m 0s
Devops part of the churn system - 2022BCS0001 sub...
CI/CD Pipeline #1: Commit 28535c9 pushed by Vishnu0107
main
Mar 21, 11:55 AM GMT+5:30
49s


Summary
All jobs
build-test-deploy
Run details
Usage
Workflow file
build-test-deploy summary
CI/CD Pipeline Report
Build Information
Trigger: Push to main branch
Environment: Ubuntu Latest (GitHub Actions Runner)
Test Execution
Test Framework: Pytest
Status: All test cases passed successfully
API endpoint (/predict) validated
Business rule logic verified
Containerization
Docker image built successfully
Image tagged using DockerHub repository
Deployment Readiness
Image pushed to DockerHub
Ready for containerized deployment
Overall Status
Pipeline completed successfully with all stages executed without errors.
Job summary generated at run-time

Summary

All jobs

build-test-deploy

Run details

Usage

Workflow file

build-test-deploy

succeeded 19 minutes ago in 58s

Search logs

Set up job 1s

Checkout Code 2s

Set up Python 0s

```
1  ▶ Run actions/setup-python@v5
9  ▶ Installed versions
```

Install dependencies 10s

Set environment variables 0s

```
1  ▶ Run echo "DATABASE_URL=sqlite:///./test.db" >> $GITHUB_ENV
```

Run Tests 1s

```
1  ▶ Run pytest --maxfail=1 --disable-warnings -q --tb=short | tee pytest_output.txt
13 ......                                                                  [100%]
14 6 passed in 0.73s
```

Upload Test Results 1s

```
1  ▶ Run actions/upload-artifact@v4
18 With the provided path, there will be 1 file uploaded
19 Artifact name is valid!
20 Root directory input is valid!
21 Beginning upload of artifact content to blob storage
22 Uploaded bytes 101
23 Finished uploading artifact content to blob storage!
24 SHA256 digest of uploaded artifact zip is 1e39577b98bb08e80db5c12b2d791af3df354808ccd84adaf4b9deb37ec64070
25 Finalizing artifact upload
26 Artifact pytest-results.zip successfully finalized. Artifact ID 6041108706
```

Set environment variables 0s

Run Tests 1s

Upload Test Results 1s

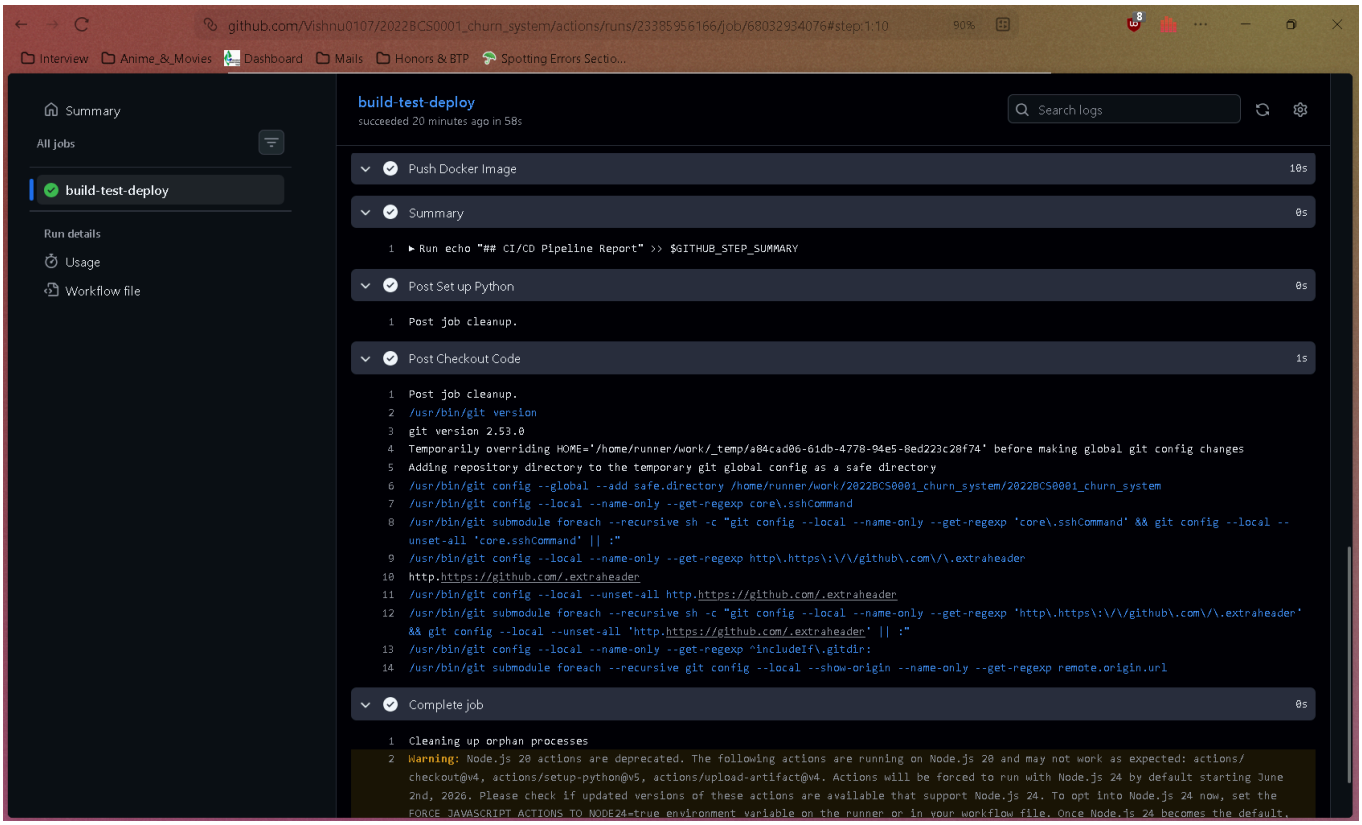
Build Docker Image 29s

Login to DockerHub 0s

```
1  ▶ Run echo "***" | docker login -u "***" --password-stdin
13
14 Login Succeeded
15 WARNING! Your credentials are stored unencrypted in '/home/runner/.docker/config.json'.
16 Configure a credential helper to remove this warning. See
17 https://docs.docker.com/go/credential-store/
18
```

Push Docker Image 10s

```
1  ▶ Run docker push ***/churn-api:latest
13 The push refers to repository [docker.io/***/churn-api]
14 b21796fa838d: Preparing
15 45dc2d2df494: Preparing
16 0567a9804dec: Preparing
17 c133d2fe7445: Preparing
18 7fe682552c2a: Preparing
19 bc7d147c4fff: Preparing
20 6afcfc3ecd04: Preparing
21 817e939a94eb: Preparing
22 dd6e353abeff: Preparing
23 c5864b4cf4c9: Preparing
24 bc7d147c4fff: Waiting
25 6afcfc3ecd04: Waiting
26 dd6e353abeff: Waiting
```


github.com/Vishnu0107/2022BCS0001_churn_system/actions/runs/23385956166/job/68032934076#step:1:10
Interview
Anime_&_Movies
Dashboard
Mails
Honors & BTP
Spotting Errors Sectio...
Summary
All jobs
build-test-deploy
Run details
Usage
Workflow file
build-test-deploy
succeeded 20 minutes ago in 58s
Search logs
Push Docker Image 10s
Summary 0s
1 ▶ Run echo "## CI/CD Pipeline Report" >> $GITHUB_STEP_SUMMARY
Post Set up Python 0s
1 Post job cleanup.
Post Checkout Code 1s
1 Post job cleanup.
2 /usr/bin/git version
3 git version 2.53.0
4 Temporarily overriding HOME='/home/runner/work/_temp/a84cad06-61db-4778-94e5-8ed223c28f74' before making global git config changes
5 Adding repository directory to the temporary git global config as a safe directory
6 /usr/bin/git config --global --add safe.directory /home/runner/work/2022BCS0001_churn_system/2022BCS0001_churn_system
7 /usr/bin/git config --local --name-only --get-regexp core\.sshCommand
8 /usr/bin/git submodule foreach --recursive sh -c "git config --local --name-only --get-regexp 'core\.sshCommand' && git config --local --unset-all 'core.sshCommand' || :"
9 /usr/bin/git config --local --name-only --get-regexp http\.https\:\/\/github\.com\/\.extraheader
10 http.https://github.com/.extraheader
11 /usr/bin/git config --local --unset-all http.https://github.com/.extraheader
12 /usr/bin/git submodule foreach --recursive sh -c "git config --local --name-only --get-regexp 'http\.https\:\/\/github\.com\/\.extraheader' && git config --local --unset-all 'http.https://github.com/.extraheader' || :"
13 /usr/bin/git config --local --name-only --get-regexp ^includeIf\.gitdir:
14 /usr/bin/git submodule foreach --recursive git config --local --show-origin --name-only --get-regexp remote.origin.url
Complete job 0s
1 Cleaning up orphan processes
2 Warning: Node.js 20 actions are deprecated. The following actions are running on Node.js 20 and may not work as expected: actions/checkout@v4, actions/setup-python@v5, actions/upload-artifact@v4. Actions will be forced to run with Node.js 24 by default starting June 2nd, 2026. Please check if updated versions of these actions are available that support Node.js 24. To opt into Node.js 24 now, set the FORCE_JAVASCRIPT_ACTIONS_TO_NODE24=true environment variable on the runner or in your workflow file. Once Node.js 24 becomes the default,